Name:	Class:	Date given:
		Date due in:

The Carbon Cycle

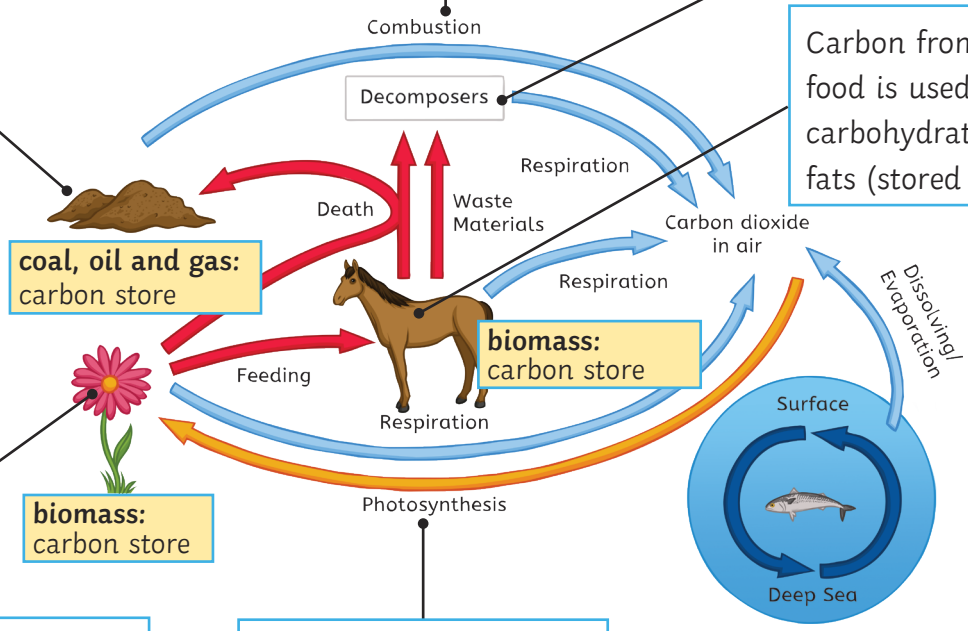
Use the carbon cycle diagram to help you answer the questions.

When they are burned to produce electricity, fossil fuels release carbon dioxide and other greenhouse gases back into the atmosphere.

Combustion engines in petrol and diesel vehicles release carbon dioxide gas into the atmosphere when used.

Waste materials and dead organisms are broken down and the carbon elements are returned to the soil or air as other compounds.

Carbon from digested food is used to make carbohydrates, proteins and fats (stored as biomass).



coal, oil and gas: carbon store

biomass: carbon store

biomass: carbon store

ocean: carbon store (sink)

Carbon from carbon dioxide gas is used in photosynthesis to make carbohydrates, proteins and fats (stored as biomass).

Deforestation (cutting down woodland) reduces the amount of carbon dioxide absorbed from the atmosphere for photosynthesis.

1. How is carbon used by living things? [2]

2. Name two stores of carbon compounds. [2]

3. Name two processes that return CO_2 to the atmosphere. [2]

4. Name a process that removes CO_2 from the atmosphere. [1]

5. Describe a human activity that affects CO_2 levels in the atmosphere. [2]

Learning Outcomes (tick if achieved)

Q1	I know what carbon is used for in living things.	
Q2,3,4	I can identify processes that move carbon in the environment.	
Q5	I can describe how humans affect the carbon cycle.	

The Carbon Cycle **Answers**

1. How is carbon used by living things? [2]

   Carbon is used by plants (producers) [1] to make carbohydrates, proteins and fats [1]

2. Name two stores of carbon compounds. [2]

   Two from: fossil fuels, biomass, carbohydrates, food

3. Name two processes that return CO_2 to the atmosphere. [2]

   Two from: combustion, respiration, dissolving/evaporation

4. Name a process that removes CO_2 from the atmosphere. [1]

   Photosynthesis

5. Describe a human activity that affects CO_2 levels in the atmosphere. [2]

   burning fossil fuels (for electricity) – increase

   deforestation (fewer plants to absorb CO_2) – increase

   driving cars (fumes and emissions) - increases

   1 mark for naming the human activity

   1 mark for describing the effect i.e. increase or decrease in CO_2 levels